

METHOD REFERENCE · TCPS-PA v3.0



Event-Centric · Dynamic Timing Contract · Bidirectional Diagnosis · Physical Budget Loss



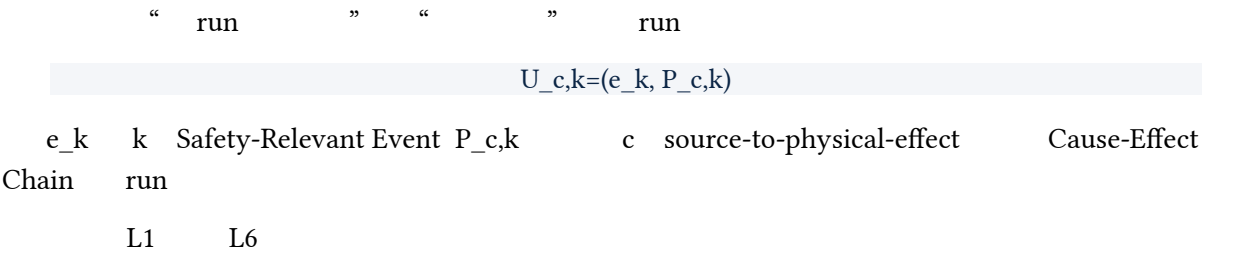
CARLA 0.9.15 + Apollo 10.0.0 + Bridge

run

Observed / Requirement / Model / Retrospective

2026-08-13

run



L1 Temporal Stressor / Root-Cause		
L2 Local Temporal Degradation		task edge sample-and-hold
L3 Cause-Effect Timing		Reaction Age Fresh Gap
L4 Dynamic Temporal Contract		miss
L5 Physical Budget Loss		
L6 Physical Safety State		

L4/L6 seed -> L3 lineage -> L2 local symptom -> L1 root-cause hypotheses
L4 contract state -> L5 physical budget loss -> L6 physical outcome

L1 verified intervention -> L2 -> L3 -> L4 -> L5 -> L6 -> attribution

-
1. chain task/edge/actuation segment

2. execution queue activation/phase transport staleness reuse gap overwrite/drop actuation / /

3. observed miss guarantee loss observed miss

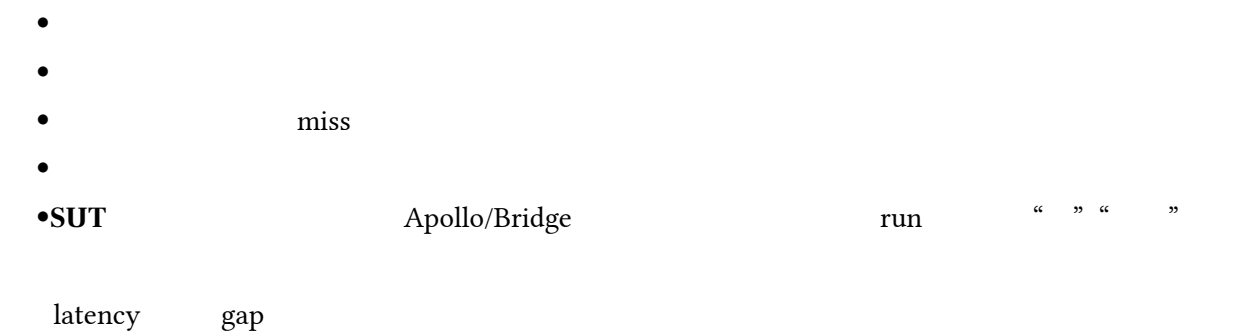
4.

5. deadline-excess distance debt

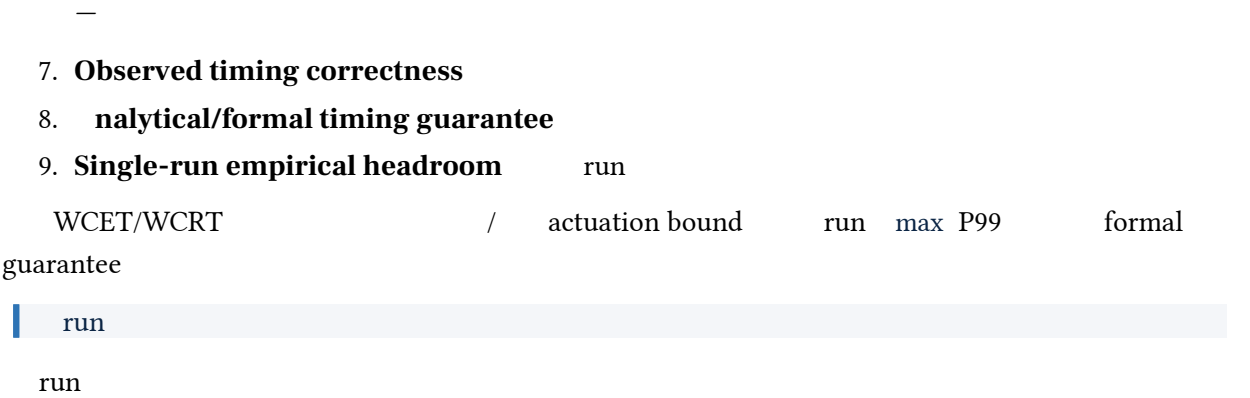
6. SUT

1.

1.1 “ ”
“ ”

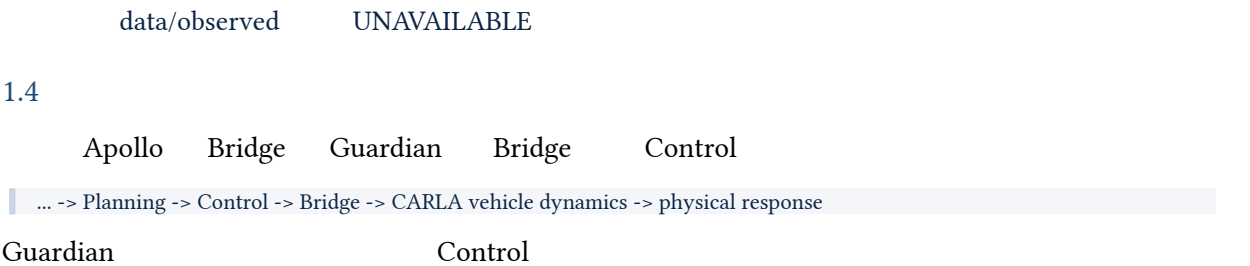


1.2



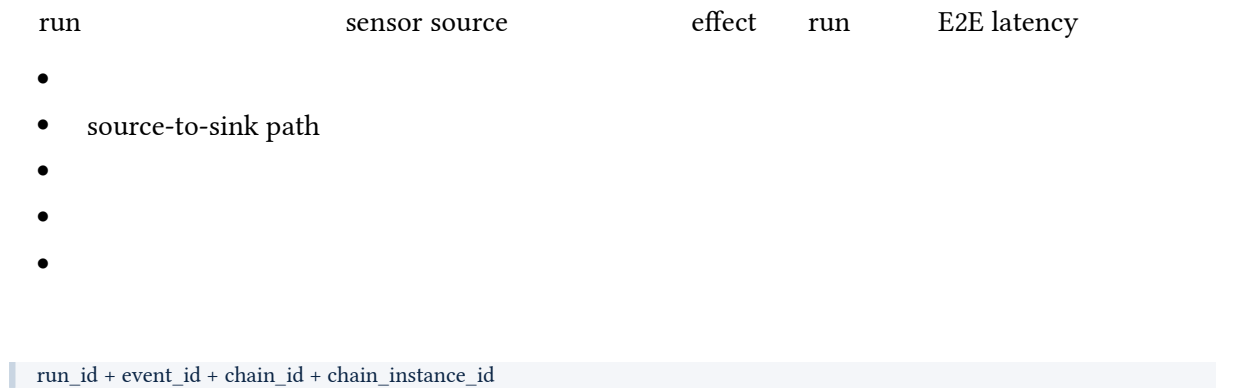
1.3

data/observed		
requirement/qualified		
model/predicted		
retrospective/reconstructed	run	

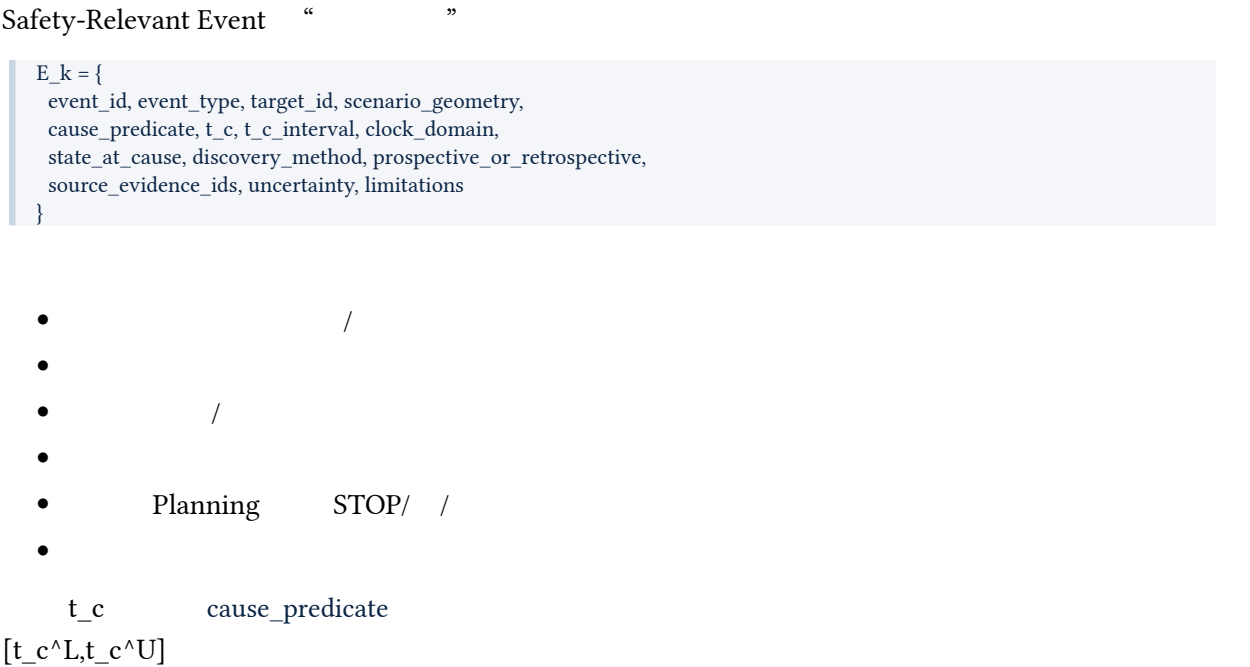


2. Run Safety-Relevant Event Cause-Effect Chain

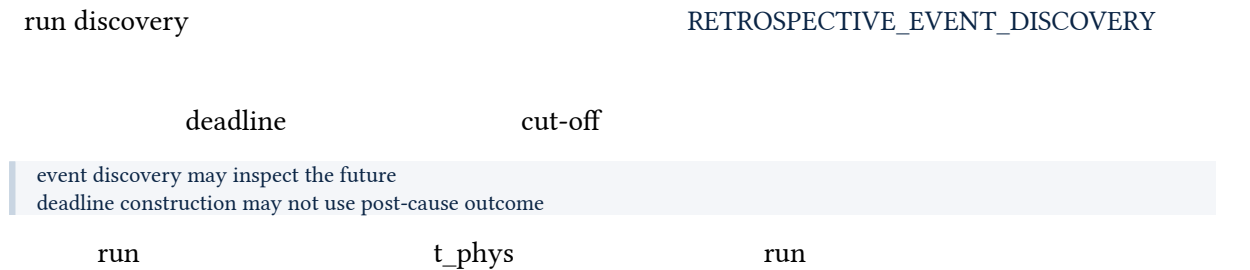
2.1 Run



2.2 Safety-Relevant Event



2.3 deadline



2.4 Cause-Effect Chain

DAG source-to-sink path source latency

LiDAR -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
 Radar -> Fusion -> Prediction -> Planning -> Control -> Bridge -> physical effect
 Localization -> Planning -> Control -> Bridge -> physical effect
 Chassis -> Control -> Bridge -> physical effect

ECRTS 2023 source/sink DAG source-to-sink cause-effect path Yi deadline
 sensor-to-actuator path task E2E deadline [ECRTS 2023](#) [Yi et al. 2021](#)

2.5 effect

“ ” effect effect predicate

Psi_effect(event_type, target_id, vehicle_state, command_state) = TRUE/FALSE

effect

- t_cmd Control
- t_apply Bridge /
- t_phys effect predicate
- effect_hold_rule

effect E2E physical Reaction Time

3.

3.1

Bridge/SCB	delay	deadline miss
Apollo	fallback solver	
Trace events	read/start/end/pub / monotonic	/
Trace message context	trace_id parent input/output seq source data timestamp	target outcome
Parsed Cyber record	channel /record receipt header/source timestamp payload reuse	read/start/finish
Localization/Chassis		CARLA sim time
Collision event	actor impulse	
Actor history/ground	CARLA frame/sim	Apollo

truth		

ApolloRecordCyber RT channel / record timeline

[Apollo 10 Cyber RT Developer Tools](#)

3.2

timestamp clock_domain host timestamp_type unit resolution alignment_method error_bound

- wall_epoch epoch
- monotonic monotonic trace
- record_time / record
- header_timestamp header
- measurement/source_time
- CARLA_sim_time/frame
- Bridge_apply_wall Bridge

10.
11.
- 12.offset/drift/anchor
13. age
14.

P_CLOCK “ ” t_c t_phys Orin

3.3 phase

“ timestamp ” phase “producer consumer CARLA tick / ”

phase quantization phase offset

- clock_alignment_audit.csv
- phase_audit.csv

phase latency PHASE_EFFECT_HYPOTHESIS phase

3.4

DIRECT_OBSERVED
OBSERVED_DERIVED
TRACE_LINEAGE

WEAK_TEMPORAL_ALIGNMENT
INDEPENDENT_REQUIREMENT
INDEPENDENT_CALIBRATED_MODEL
VALIDATED_MODEL
UNVALIDATED_MODEL
RETROSPECTIVE_RECONSTRUCTION
OUTCOME_TRUNCATED
CONFLICTED_EVIDENCE
MISSING

CLOCK_TAINT TARGET_TAINT MODEL_TAINT RETRO_TAINT OUTCOME_TRUNCATED

3.5

•P_TARGET sensor/Fusion/Prediction/Planning/Control/ actor
•P_FUNC Perception Prediction Planning trajectory
Control trajectory Bridge
P_FUNC=QUALIFIED_PASS deadline “ ” Planning STOP trajectory
infeasible Control

4. L3 Cause-Effect Timing

run L3/L4 L6 seed L1

4.1

trace record Localization Bridge

```
event_time.csv
{
  event_uid, run_id, event_id, chain_id, module, stage,
  event_kind, read_or_write, clock_domain, host,
  raw_time, aligned_wall_time, alignment_error,
  input_seq, output_seq, trace_id, parent_trace_id,
  data_source_time, target_id, payload_digest,
  source_file, source_locator, evidence_class, quality_flags
}
```

event_kind physical_cause sensor_sample receive/read ready start end publish/write command
bridge_apply physical_effect outcome

4.2

Grade		
A	trace_id parent/provenance input/output seq	cause-effect propagation
B	source/header timestamp + downstream mapping	propagation

Grade		
C		/
D	nearest-time heuristic	
UNKNO WN		

P99 stage decomposition causal chain instance

4.3 Forward causal tracing

source sample “ source ” job

source sample -> earliest eligible downstream read -> ... -> first qualifying command -> physical effect

- source data sink
- consumer
-
- /
- Control Bridge

4.4 Backward provenance tracing

effect Control Planning

physical effect <- applied command <- Control input trajectory
<- Planning inputs <- Prediction/Fusion source samples

“ ” Data Age reuse/staleness Forward backward

- forward sample effect
- observed effect sample
- overwrite skip reuse

4.5 ECRTS partition

ECRTS 2023 p-partitioned job chain backward chain forward chain warm-up
MRT MDA partition point

at partition p:
backward provenance from p to source
+ forward propagation from p to physical effect

-
- read/write job warm-up/valid-chain job-chain

- PARTITION_DIAGNOSTIC_VIEW MRT/MDA

4.6 Multi-input Age Vector

Planning/Control

Age

$$A_{c,k}(t_e) = [A_{\text{perception}}, A_{\text{prediction}}, A_{\text{localization}}, A_{\text{chassis}}, A_{\text{map}}, \dots]$$

$$A_j = t_{\text{effect}} / \text{read-}t_{\text{source},j}$$

source semantics chain target relevance clock error A_{critical}
vector critical source

4.7 Fresh Causal Update Gap

output gap

effect

q_n

qualifying(q_n) iff
target/event semantics match
AND source timestamp strictly advances
AND lineage reaches the selected sink/effect
AND output is not mere reuse of the same source state

$$G^{\text{fresh}}_n = t(q_{n+1}) - t(q_n)$$

• G_{publish}

• $G_{\text{source_advance}}$ source timestamp

• $G_{\text{fresh_effect}}$ sink/effect

“Control 100 Hz ” “ 10 ms ” Control Planning trajectory

5. L2

5.1 task/edge

producer i-1 consumer i

$t_{\text{write_prev}}$ producer
 t_{receive} consumer/record
 t_{ready} job
 t_{start} job
 t_{end} job
 t_{write} consumer

$$L_i^{\text{transport}} = t_{\text{receive}} - t_{\text{write_prev}}$$

$$L_i^{queue} = t_{ready} - t_{receive}, L_i^{activation} = t_{start} - t_{ready}$$
$$L_i^{exec} = t_{end} - t_{start}, L_i^{publish} = t_{write} - t_{end}$$

ready start-receive queue phase unresolved_wait record

record

5.2

Sampling delay	t_s-t_c	sensor tick phase
Transport delay	receive - producer write	SHM/RTPS bridge
Queue/backlog	ready - receive /	CPU/GPU contention callback backlog
Activation/phase wait	start - ready	timer/callback phase
Execution tail	end - start	fallback
Publish delay	write - end	writer/backpressure
Staleness	consumer time - source time	producer
Reuse	sink source/trajectory	sample-and-hold gap
Overwrite/skip	source update downstream successor	buffer policy
Drop/reorder	seq /	transport/queue policy
Actuation delay	physical effect - applied command	Bridge

5.3 R/A/G

$$T_{deg} = [R, A, G]$$

- R reaction/response-time behavior
- A data freshness/age
- G update continuity/gap

run count P50 P90 P95 P99 MAX IQR/MAD run run

formal bound

5.4 run “ ”

L2 “ ” reference run

- NOMINAL_PERIOD
- ENGINEERING_REQUIREMENT /edge timing requirement

- PRE_FAULT_WITHIN_RUN run
 - DECLARED_EXTERNAL_REFERENCE
 - LOCAL_STATIONARY_REFERENCE run
- context covariates defeater

reference

run 412 ms Planning

Planning continuity degraded

5.5

Prediction -> Planning 85 ms edge/segment phase queue scheduler interference

ready/start / L2

observed symptom
 candidate mechanisms
 supporting evidence
 challenging evidence
 missing discriminator

6. L1 Temporal Stressor

6.1

Discovery mode L1 L4 miss L6 loss L3/L2 anomaly

Validation mode L1 “ delay”

delay

6.2 Temporal Fault/Stress Signature

```
F_T = {
  fault_or_stressor_type, location, onset, end, duration,
  requested_magnitude, actual_magnitude, distribution,
  persistent_or_transient, one_shot_or_repeated,
  affected_channels/messages, queue_policy,
  drop_status, reorder_status, resource_context,
  direct_evidence, confidence
}
```

- sampling/trigger delay
- execution/WCET tail
- CPU/GPU/memory/lock contention

- scheduler/priority/blocking
- queue/backlog/backpressure
- network/transport/Bridge delay
- period/phase/mode-transition delay
- staleness/reuse/overwrite/drop/reorder
- actuation/control-build-up delay
- clock/timestamp artifact
- functional or physical alternative

6.3

“ ” applied row Bridge/SCB Bridge/Control
 Apollo
 fault onset t_c $[t_f, t_c]$ pre-hazard state divergence /

7. L4 Per-Event / Per-Chain Dynamic Temporal Contract

L4

7.1 deadline

x_c ρ / H
 $\Phi(x_c, \rho) = \min_{t \in [0, H]} h(x_{\text{ego}}(t), x_{\text{target}}(t), \text{geometry}, \text{policy})$
 $\Phi \geq 0$ dynamics safety buffer deadline
 $\tau_{R,c,k}^{\text{req}} = \sup\{\rho \geq 0 \mid \exists f \in \Theta_m \Phi(x_c, \rho) \geq 0\}$
 Θ_m context/ ODD deadline x_c

7.2 /

RSS-like
 $d_{\text{req}}(\rho) = v_{\text{ep}} + (1/(2)a_{\text{respp}}^2 + \frac{(v_{\text{e}} + a_{\text{respp}})^2}{2b_{\text{e}}} - (v_{\text{f}}^2)/(2b_{\text{f}}) + d_{\text{safe}})$
 $d_{\text{req}}(\rho) \leq d_{\text{clear}}$ ρ $v_{\text{f}} = 0$ $\rho = 0$
 $\tau_{\text{req}} = 0$
 $\text{state} = \text{ALREADY_OUTSIDE_DECLARED_ENVELOPE}$
 delay

Koopman Osyk Weast longitudinal model response period
 braking trajectory closest approach [Koopman et al. 2019](#)

reachability/trajectory model

deadline

NOT_QUALIFIED_PRIMARY

7.3 Deadline

deadline observed miss

- state input t_c cut-off
- target identity geometry target-motion assumption
- / dynamics braking envelope
- calibration runs evaluation run
- friction grade curvature actuation build-up uncertainty bounds
- run post- t_c t_{phys}
- $\tau_{req_low/center/high}$ provenance

UNVALIDATED_MODEL RETROSPECTIVE_RECONSTRUCTION

NOT_QUALIFIED_PRIMARY

7.4 ODD / context uncertainty

context region m

$$\tau_{robust}(m) = \epsilon \{x \in X_m \mid \tau(x)\}$$

“ region ” ODD “region ” “ region ”

[Koopman et al. 2019](#)

7.5

Reaction Freshness Fresh Update Continuity

$$S_{obs} = [\tau_R^{req} - T_R^{obs}; \tau_A^{req} - A_{critical}^{obs}; \tau_G^{req} - G_{fresh}^{obs}]$$

 $\tau_A^{req} \quad \tau_G^{req} \quad \tau_R^{req}$

- Freshness contract “ a / ”
- Fresh-update contract “ g / reachable states ”
- A/G freshness/gap requirement reaction deadline

ECRTS warm-up MRT MDA job-chain observed Reaction Age

A/G [ECRTS 2023](#)

7.6 observed contract verdict

 $[T_L, T_U]$ deadline $[\tau_L, \tau_U]$

$T_U \leq \tau_L$	CLEARLY_WITHIN_CONTRACT

$T_L > \tau_U$	CLEARLY_MISSED
	BOUNDARY_UNCERTAIN
deadline	RETROSPECTIVE_ONLY
	MODEL_SUPPORTED_ONLY
/clock/target	NOT_TESTABLE

observed slack

7.7 Observed correctness Timing Guarantee

$$S_{\text{obs}} = \tau_R^{\text{req}} - T_R^{\text{obs}}$$

$$S_{\text{guar}} = \tau_{R,\text{low}}^{\text{req}} - R_{\text{path}}^{\text{UB}}$$

$R_{\text{path}}^{\text{UB}}$ / / WCET/WCRT release/period priority/scheduler
blocking queue/transport multi-rate overwrite semantics phase actuation bound

Yi $\Sigma_i(p_i + r_i)$ $2\Sigma_i p_i$ CPU EDF WCET Apollo
budget Apollo trace period formal guarantee [Yi et al. 2021](#)

$$S_{\text{obs}} \geq 0 \quad S_{\text{guar}} < 0$$

$$R^{\text{UB}} \quad S_{\text{guar}} = \text{NOT_ESTABLISHED}$$

single_run_empirical_headroom = tau_req - max observed comparable instance

guarantee

7.8 Dynamic deadline shrink mode transition

Yi deadline deadline job sensor data
E2E deadline [Yi et al. 2021](#)

hazard window contract trajectory

$$d/dt S(t) = d/dt \tau^{\text{req}}(t) - d/dt T^{\text{risk}}(t)$$

$$S > 0, d/dt S < 0$$

$$T_{\text{exhaust}}^{\text{local}} = (S) / (-d/dt S)$$

deadline region/mode change job trajectory

contract

7.9 Slack Waterfall Guarantee-Loss Point

chain i

$$L_{\text{prefix},i}^{\text{obs}} = t_i - t_c$$

$$B_i^{\text{obs}} = \tau_R^{\text{req}} - L_{\text{prefix},i}^{\text{obs}}$$

i effect suffix bound $R_i \rightarrow \text{phys}^{\text{UB}}$

$$B_i^{\text{guar}} = \tau_{R,\text{low}}^{\text{req}} - L_{\text{prefix},i}^{\text{obs}}, U - R_i \rightarrow \text{phys}^{\text{UB}}$$

sampling transport queue/unresolved wait activation execution publish consumer wait
command-to-apply actuation

15. FIRST_CONDITIONAL_GUARANTEE_LOSS $B_i^{\text{guar}} < 0$

16. FIRST_IRREVERSIBLE_OBSERVED_MISS effect τ_U miss

17. OBSERVED_EFFECT_MISS t_{phys} miss

FIRST_CONDITIONAL_GUARANTEE_LOSS “ + ”

segment budget/reference

suffix bound observed slack waterfall guarantee-loss point

8. seed

8.1 seed

18. L4 contract miss

19. L6 L4

20. L3 physical Reaction/Age/Fresh Gap

21. L2 reference

22. reference case-level anomaly

seed C1

8.2 Backward diagnosis

seed

23. **seed** chain outcome

24. **lineage segment** effect/command source budget

25. sampling execution queue activation/phase transport staleness/reuse overwrite/
drop mode transition Bridge/actuation

- 26. clock artifact target mismatch functional failure initial unsafe state braking/road/geometry outcome conflict
- 27. onset seed
- 28. supporting challenging/contradicting missing evidence
- 29. equivalence class
- 30. discriminator trace /
- 31. /

8.3

SUPPORTED_CANDIDATE
CONSISTENT_BUT_UNRESOLVED
CHALLENGED
REFUTED
NOT_TESTABLE
NOT_APPLICABLE

DETECTED	/
LOCALIZED_TO_SEGMENT	
MECHANISM_SUPPORTED	
ISOLATED_CANDIDATE	
INTRINSIC_DEFECT_SUPPORTED	/ /

run “ ” LOCALIZED_TO_SEGMENT ISOLATED_CANDIDATE

8.4 Forward validation

verified stressor
-> matching local primitive changes
-> causally matched chain timing effect
-> qualified contract state
-> physical budget loss
-> outcome

backward candidate

9. L5 Physical Safety Budget Loss

9.1 Canonical observed response distance

cause t_c effect t_{phys}

$$D_{response,wall}^{data/observed} = \int_{t_c}^{t_{phys}} v(t) dt_{wall}$$

$D_{response_wall_integral_data_observed_m}$

$D_{delay_wall_integral_data_observed_m}$ CARLA frame/sim time Localization
realtime factor

9.2 Deadline-excess distance debt

deadline endpoint

$$t_d = t_c + \tau_R^{req}$$

miss

$$D_{debt}^{requirement \setminus constrained} = \max_{\leq} (0, \int_{t_d}^{t_{phys}} v(t) dt_{wall})$$

“ ” ancestry “ requirement + observed velocity path”

REQUIREMENT_CONSTRAINED_DERIVED

- model deadline -> $D_{debt_model_predicted_m}$
- retrospective deadline -> $D_{debt_retro_diagnostic_m}$
- deadline -> D_{debt}

9.3

$$M_0^{data/observed} = D_1 - D_{response}^{obs} - D_{brake}^{obs}$$

$$M_{safe}^{data/observed} = M_0^{obs} - D_{safe}$$

D_1 $D_{response}$ D_{brake} fault onset t_c

• Total Closed-loop Effect t_f outcome

• Post-cause Response Effect t_c

response distance deadline-excess debt

9.4

/

$$M_{phys}^{data/observed} = \min_{t \in [t_c, t_o]} h_{observed}(x_e(t), x_o(t), geometry)$$

CARLA ground truth/actor history outcome Apollo “ ”

$$M_{\text{phys}}^{\text{model/predicted}}(\rho) = \min_t h_{\text{model}}(x_e(t; \rho), x_o(t), \Theta_m)$$

slack margin margin margin

9.5 “ ”

run

32.D_response_wall_integral_data_observed_m

33.D_debt_requirement_constrained_m deadline

34.M_phys_data_observed_m / outcome

35.Delta_M_phys_model_predicted_m ρ_{ref}

$$\Delta M_{\text{phys}}^{\text{model}} = M_{\text{phys}}^{\text{model}}(\rho_{\text{ref}}) - M_{\text{phys}}^{\text{model}}(T_R^{\text{obs}})$$

4 “ timing X ”

9.6 Collision right-censoring

run

•collision actor time/frame impact speed/impulse

• observed response/braking segment

• actual clearance/contact state

“ observed braking distance full-stop margin”

UNAVAILABLE_OUTCOME_TRUNCATED observed

10. L6 Physical Safety State

L6

•minimum/final clearance

•physical/contact/engineering margin

•minimum speed strict stop near stop

•impact speed impulse collision actor

•truncated braking time/distance

•safety envelope violation duration/area

outcome

SAFE
SAFE_LOW_MARGIN
CRITICAL_MARGIN
MINIMUM_DISTANCE_VIOLATION
EMERGENCY_MANEUVER
NEAR_MISS
CONTACT
COLLISION

HIGH_SEVERITY_COLLISION
UNCERTAIN

CONTACT/COLLISION low/critical/near/high-severity provenance

/

11. run

Step 0

- run inventory collect window trace collision actor history record Bridge/SCB
- run analysis directory
- run record run record

Step 1

- profile parse error channel
- timestamp dictionary clock alignment phase audit
- NOT_TESTABLE “ ”

Step 2

- conflict predicate margin local minimum contact/collision candidate events
- candidate t_c target geometry effect predicate discovery provenance
- / candidate IDs

Step 3

- P_TARGET P_FUNC
- fallback/infeasibility
- Guardian Bridge

Step 4 paths

- DAG/config/trace/record channel source-to-effect paths
- path forward causal tracing backward provenance
- lineage grade

Step 5 L3 observed chain metrics

- T_sampling=t_s-t_c
- T_software=t_cmd-t_s
- T_actuation=t_phys-t_cmd command-to-apply + apply-to-physical

- $T_R = t_{\text{phys}} - t_c$
- Age Vector publish/source-advance/fresh-effect gaps
- uncertainty

Step 6 L2

- chain instance transport/wait/activation/exec/publish/reuse
- -run reference
- reference
- local symptom

Step 7 L4 contract

- safety model
- cut-off state dynamics/uncertainty
- τ_R τ_A τ_G low/center/high
- anti-leakage domain qualification
- model/retrospective observed contract verdict

Step 8 observed correctness guarantee

- uncertainty-aware contract verdict
- suffix/WCRT bounds slack waterfall guarantee-loss points
- formal bound observed waterfall single-run empirical headroom
- shrinking contract mode-transition/old-job/old-trajectory

Step 9 Backward diagnosis L1

- miss/loss seed segment
- hypothesis ledger equivalence classes defeaters discriminators
-

Step 10 L5/L6

- observed wall-clock velocity D_{response}
- deadline D_{debt}
- margin/outcome
- observed
- collision run right-censoring

Step 11

- clock phase target function initial state braking geometry freshness outcome conflict defeaters

- Evidence Ledger -> Claim Ledger -> Defeater Ledger
-

12.

12.1

- Perception target / tracking continuity
- Prediction target/trajectory
- Planning STOP/decision/fallback/infeasibility/latency debug
- Control command/latency
- Localization speed/acceleration/pose
- Bridge/SCB requested/applied delay
- collision and actor history

timestamp

12.2 Trace

e2e_trace_v3

```
events: event_id, trace_id, module, phase, mono_ns, pid, tid
message_context: edge, channel, input_seq, output_seq,
                 data_ts_ns, primary_parent_trace_id,
                 parent_trace_count, trace_valid
```

- Grade A/B lineage
- proc/runonce/substage timing
- source timestamp
- parent/fusion lineage
- overwrite/skip

monotonic trace trace anchor uncertainty

12.3 Parsed Cyber record

record

- record_message_index channel coverage/rate/gap
- Localization/Chassis state
- Planning header task latency decision trajectory
- Control command latency vehicle state Planning reuse/age

- Perception/Prediction obstacles source headers
 - sensor timeline
 - Planning-Control alignment sensor-to-Control message diagnostics
-
- channel protobuf
 - record_time / task start
 - sensor_to_control_reaction_time source-to-Control t_c t_phys physical Reaction Time
 - obstacle table obstacle row gap channel timeline
 - Planning total_time_ms task row frame statistic planning_seq
 - run record
 - run record
 - record-only Age/reuse/message payload
 - trace/log
 - L3 lineage L2 mechanism

12.4 Collision

- collision event/JSONL outcome
- actor history realtime factor
- Localization wall-time velocity
- CARLA frame/sim time e2e_sim_frames realtime_factor

13.

```

analysis/<run_id>/
├── report/
│   └── single_run_realtime_diagnosis.md
├── tables/
│   ├── input_inventory.csv
│   ├── event_catalog.csv
│   ├── event_timeline.csv
│   ├── chain_catalog.csv
│   ├── chain_instances.csv
│   ├── partition_lineage.csv
│   ├── local_timing_primitives.csv
│   ├── age_vector.csv
│   ├── fresh_update_gaps.csv
│   ├── temporal_fault_signature.csv
│   ├── clock_alignment_audit.csv
│   ├── phase_audit.csv
│   ├── target_identity_audit.csv
│   ├── functional_correctness_audit.csv
│   ├── requirement_registry.csv
│   └── dynamic_contract_construction.csv

```

```

├── contract_verdicts.csv
├── slack_waterfall.csv
├── guarantee_loss_points.csv
├── velocity_trajectory_observed.csv
├── physical_budget_observed.csv
├── physical_budget_model_predicted.csv
├── physical_outcomes.csv
├── diagnosis_hypothesis_ledger.csv
├── diagnosis_edges.csv
├── evidence_ledger.csv
├── claim_ledger.csv
├── defeater_ledger.csv
├── exclusions_and_missing.csv
├── validation/
│   ├── data_quality_audit.md
│   ├── anti_leakage_audit.md
│   ├── arithmetic_recomputation.md
│   ├── claim_audit.md
│   └── validation.json

```

13.1

Event / Target / Geometry
 Cause predicate and t_c interval
 Selected critical chain(s) and lineage grade

 Observed chain timing
 T_{sampling} / T_{software} / $T_{\text{command_to_apply}}$ / $T_{\text{actuation}}$ / T_R
 Age Vector / $A_{\text{critical basis}}$
 G_{publish} / $G_{\text{source_advance}}$ / $G_{\text{fresh_effect}}$

Dynamic contracts
 τ_R low/center/high + qualification
 τ_A and τ_G if independently available
 observed verdict and S_{obs}
 formal/analytical guarantee status and S_{guar}
 empirical headroom, explicitly non-guarantee

Slack waterfall
 per-segment observed budget use
 first conditional guarantee-loss point
 first irreversible observed miss
 dominant over-budget segment relative to declared reference

Diagnosis
 localized segment
 mechanism candidates and evidence
 observational equivalence class
 competing functional/clock/phase/physical explanations
 diagnosis strength and missing discriminator

Physical propagation
 $D_{\text{response_wall_integral_data_observed_m}}$
 $D_{\text{debt_requirement_constrained_m}}$
 $M_{\text{phys_data_observed_m}}$
 $\Delta M_{\text{phys_model_predicted_m}}$
 right-censoring status

Outcome and allowed conclusion

14. Claim–Evidence

I-M-E-C-O-N

- I / Inputs
- M / Metrics
- E / Evidence /
- C / Criterion
- O / Output verdict confidence ceiling
- N / Next

14.1

ID	
C1	stressor/
C2	reference
C3	/ effect
C4	observed timing
C5	timing violation
C6	/
C7	

C7

14.2

Level	
0	/
1	reference
2	cause-effect propagation
3	requirement observed contract miss guarantee loss
4	timing
5	timing-dominated safety-failure candidate
6	SUT-intrinsic temporal defect

/

14.3 defeaters

clock alignment uncertainty
 phase/tick quantization
 target mismatch
 functional failure or fallback
 initial unsafe state
 pre-cause treatment-induced state divergence
 braking capability / friction / grade / curvature mismatch
 geometry or actor association conflict
 record/trace window incomplete
 outcome truncation
 deadline leakage or model-domain mismatch
 freshness/reuse alternative
 external injected fault versus intrinsic Apollo defect

defeater OPEN/UNKNOWN UNCERTAIN/PARTIAL_PASS

15.

15.1

CHAIN_IDENTIFIED
 SEGMENT_LOCALIZED
 TASK_OR_EDGE_LOCALIZED
 MECHANISM_SUPPORTED
 ROOT_CAUSE_ISOLATED
 NOT_LOCALIZABLE

“ task” /reference over-consumption

15.2

SAMPLING
 TRANSPORT
 QUEUE_BACKLOG
 ACTIVATION_PHASE
 EXECUTION
 PUBLISH_BACKPRESSURE
 STALE_DATA
 REUSE_SAMPLE_HOLD
 OVERWRITE_SKIP
 DROP_REORDER
 MODE_TRANSITION
 BRIDGE_DELAY
 ACTUATION
 CLOCK_ARTIFACT
 FUNCTIONAL_ALTERNATIVE
 PHYSICAL_ALTERNATIVE
 UNRESOLVED_MIXTURE

15.3

- observed_contract_miss_time
- first_conditional_guarantee_loss_boundary/time
- first_irreversible_observed_miss_time
- contract_shrink_crossing_time
- formal bound GUARANTEE_NOT_ASSESSABLE

15.4

supported mechanism
 + causal segment
 + consumed budget relative to reference
 + evidence IDs
 + counter-evidence
 + unresolved equivalent candidates
 + missing discriminator

15.5

provenance

$$L_{\text{physical}} = (D_{\text{response}}^{\text{obs}}, D_{\text{debt}}^{\text{req}}, M_{\text{phys}}^{\text{obs}}, \Delta M_{\text{phys}}^{\text{model}})$$

16.

- 36.collision -> deadline miss
- 37.deadline miss -> collision fallback
- 38.output exists -> function correct
- 39.record_time difference -> execution time record task start/end
- 40.Control publishes at 100 Hz -> fresh control every 10 ms Planning
- 41.MRT maximum = MDA maximum -> every RT equals every Age
warm-up/chain
- 42.p-partition point -> root cause proven /
- 43.single-run P99/max -> formal guarantee run
- 44.first guarantee-loss point -> fault origin
- 45.model deadline -> observed deadline miss
- 46.same-run stopping distance -> independent deadline
- 47.CARLA sim frame displacement -> D_response

48. collision-truncated trajectory -> full observed stopping distance
 49. Bridge injection -> Apollo intrinsic defect
 50. $D_{\text{response}} + D_{\text{debt}}$ both subtracted debt response distance

17. run

event	ground truth/target/ predicate	EVENT_UNCERTAIN
chain	trace/provenance/headers	ASSOCIATION_ONLY / NOT_TESTABLE
	read/start/end/write	UNRESOLVED_SEGMENT
miss	observed physical RT + qualified deadline	MODEL_ONLY / RETROSPECTIVE_ONLY / NOT_TESTABLE
	qualified path/suffix upper bounds	GUARANTEE_NOT_ESTABLISHED
	+ alternatives	CANDIDATE_SET_ONLY
	wall velocity + endpoints + qualified deadline	RESPONSE_DISTANCE_ONLY / UNAVAILABLE
	collision/geometry/complete trajectory	OUTCOME_UNCERTAIN
Apollo		NOT_ESTABLISHED_FROM_SINGLE_RUN

NOT_TESTABLE

run

18.

- event predicate t_c
- target
- chain causal instance
- forward backward lineage warm-up/overwrite
- Reaction Age Gap

- software command physical effect
- record/header/source/trace/wall/sim time
- clock error slack verdict

Contract

- deadline
- cut-off run outcome
- geometry/model/domain
- low/center/high
- observed correctness formal guarantee empirical headroom

- guarantee-loss point root cause
- overrun reference/budget
- phase/queue/execution discriminator
- backward hypothesis forward proof

- D_response
- D_debt deadline
- moving target closest approach
- observed model
- collision truncation

- defeaters
- claim level
-

19.

19.1 ECRTS 2023

cause-effect chain read/write job immediate forward/backward chain MRT/MDA
 partition warm-up path
 MRT=MDA Reaction=Age partition theorem root-cause localization theorem
 job read/write

19.2 Yi et al. 2021

	sensor arrival	E2E deadline	source-to-sink path constraint	task schedulability	E2E
guarantee	shrinking deadline/mode-transition				
	CPU	EDF	WCET	$2\Sigma p_i$	Apollo/Cyber RT
bound	—	deadline			

19.3 Koopman, Osyk & Weast 2019

	context-dependent safety envelope	response time	braking trajectory	closest approach
friction/slope/curvature/vehicle/environment variability	RSS-like		ODD	
	ODD			/

20.

—					
$\{ (e_k,P_c,k) \rightarrow \leq ft(T_R^{\wedge}obs, A^{\wedge}obs,G_fresh^{\wedge}obs) \parallel \leq ft(\tau_R^{\wedge}req,\tau_A^{\wedge}req,\tau_G^{\wedge}req) \}$					
$\{ S_obs=\tau^{\wedge}req- T^{\wedge}obs\}$					
$\{S_guar=\tau_R,low^{\wedge}req-R_path^{\wedge}UB\}$					
chain	observed/guarantee slack waterfall		miss	L2 primitives	L1
$\{ D_response^{\wedge}obs, D_debt^{\wedge}req, M_phys^{\wedge}obs, \Delta M_phys^{\wedge}model, Outcome^{\wedge}obs \}$					
300 ms					
e_k	P_c,k	observed Reaction/Freshness/Continuity	requirement-constrained		
distance debt					

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